

Qualifying Exam in Microeconomics
Arizona State University
June 2017

There are four questions and you have four hours to complete the exam. If you believe that a question is in error, please explain the error and continue working.

1. A consumer's preferences on the consumption set $\mathbb{R}_+^2$ are represented by the *continuous* and *quasiconcave* function

$$u(x_1, x_2) = x_2 + \sqrt{x_1 + x_2^2}.$$

A solution to later parts of this question might use facts from earlier parts. If you have trouble with earlier parts, you should *assume* the conclusion of those parts in what follows.

- a. Confirm that ~~(i)~~ the preferences that u represents are strongly monotone; and ~~(ii)~~ the indifference curves are straight lines. For ~~(ii)~~ we strongly suggest you solve for the indifference curves explicitly, writing x_2 as a function of x_1 and the fixed utility level. ~~(iii)~~ Sketch a few indifference curves.

- ~~b.~~ Confirm that the expenditure function is given by

$$e(p, \bar{u}) = \min\{p_1 \bar{u}^2, p_2 \frac{\bar{u}}{2}\}$$

where $\bar{u} > 0$ is a utility level and $p = (p_1, p_2) \gg 0$ is a price vector.

- c. Note that e is not differentiable in p_1 globally. What preference property is responsible for this lack of differentiability? (A sentence suffices.)

- ~~d.~~ Write down the Walrasian demand correspondence, $d(p, w) = (d_1(p, w), d_2(p, w))$, where $w > 0$ is the consumer's wealth and d_ℓ is the demand for good ℓ .

- ~~e.~~ Call good ℓ *normal* if for every $p \gg 0$ and $w_2 > w_1 > 0$, $x_\ell^t \in d_\ell(p, w_t)$ for $t = 1, 2$ implies $x_\ell^2 \geq x_\ell^1$. In words: good ℓ is normal if its demand *does not decrease* when wealth *increases*. Is good 1 normal? Is good 2? Explain.

really?
 $\frac{w}{p_2} \geq \sqrt{\frac{w}{p_1}}$
 $\frac{4w^2}{p_2} \geq \frac{w}{p_1}$
 $\frac{4w}{p_2} \geq \frac{p_2}{p_1}$

2. Consider a standard model of market signaling. After observing her type t , a worker chooses a level of education $e \geq 0$. The market, i.e. the second player, observes e and offers a wage w . The worker accepts or rejects the offer.

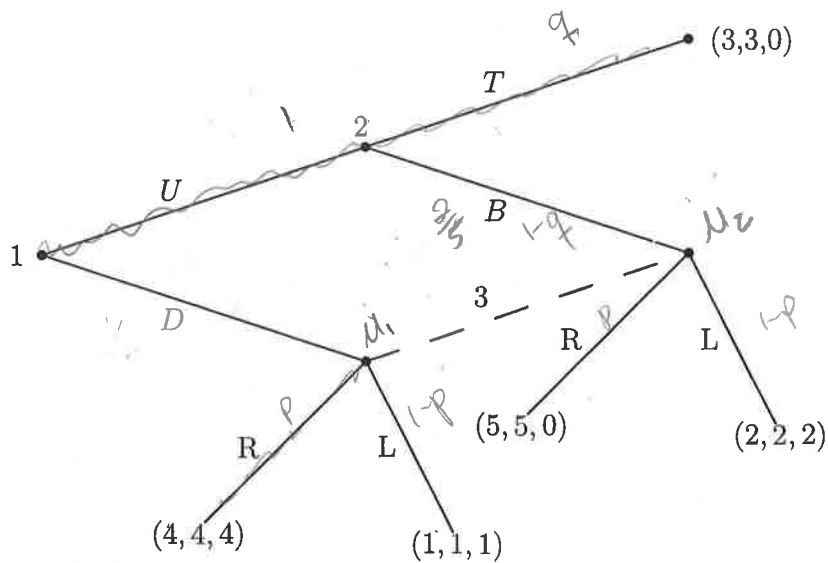
The worker's private information t is the worker's productivity. A worker's type t can take any value in $T = \{1, 2, \dots, n\}$ according to a probability distribution P such that $\forall t, P(t) > 0$. The cost to acquire education level e for a worker of type t is $f(t)e^2$ where $f(t)$ is a strictly decreasing positive function of t . The utility of a worker of type t who chooses education e and accepts a wage w is $U(t, e, w) = w - f(t)e^2$. The utility of a worker of type t who chooses education e and does not accept a wage offer is $-f(t)e^2$.

Education has no productive value for firms. The assumption of a competitive market is interpreted as a zero profit condition, that is to say, the wage offered given the available information to the market, is the expected productivity given the available information to the market.

- a. Specify the set of pure strategies for the worker. Fix an arbitrary pure strategy for the worker. What is the wage schedule offered by the "market?" *no x*
- b. Construct a separating equilibrium in which type t chooses $e(t)$. Hint: identify $e(1)$ and express $[e(t)]^2$ as a function of $[e(t-1)]^2$ to obtain a valid expression for $e(t)$. What is the role of P in your construction?
- c. Specify beliefs that would make the separating equilibrium you constructed into a Weakly Perfect Bayesian Equilibrium.

3. Consider the three-player game represented in the graph below. Note that player 1 moves first and that the dashed line indicates an information that belongs to player 3.

Find two sequential equilibria that differ in more than the beliefs supporting them. Show that they are sequential equilibria.



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4. An economy has two goods, a private good and a single public good that is either produced ($x = 1$) or not ($x = 0$). To produce the public good it takes $K > 0$ units of the private good.

There are two consumers each with expected utility preferences that are quasilinear in the private good. Specifically, consumer i 's vN-M utility is $u_i(x, y_i) = xv_i + y_i$, where y_i is i 's consumption of the private good. Consumer i is endowed with $m_i > \max\{0, v_i\}$ units of the private good, whose price is normalized to 1. The interpretation is that i values the public good at $v_i \in \mathbb{R}_+$ dollars.

Consider the following game. Each consumer submits a bid $b_i \in [0, m_i]$ in units of the private good. The bids determine whether the public good is produced, and each consumer's tax bill if produced. If the sum of the bids is less than K , then the public good is not produced and no taxes are collected. If the sum of the bids is at least K , then the public good is produced, and consumer i 's tax bill is $t_i = K - b_j$ units of the private good, $j \neq i$. In words: if produced, a consumer pays the cost K minus the bid of other consumer.

The valuations are commonly known, but the consumers do not know the cost K . Each thinks it is distributed according to a differentiable density function f that is positive on all of $\mathbb{R}_+$, and 0 otherwise, with finite mean μ_K .

- ~~a.~~ Write down consumer i 's expected utility as a function of a bid profile (b_1, b_2) .
- ~~b.~~ Identify any Nash equilibria of this game. Justify your conclusion carefully.
- c. Suppose that consumer 1 now becomes strictly risk averse: consumer 1's vN-M utility is now $v_1(x, y_1) = T(u_1(x, y_1))$ for some strictly increasing, strictly concave, and continuously differentiable function $T : \text{Range}(u_1) \rightarrow \mathbb{R}$. What happens to player 1's best reply?

bad
watching
it really have
a 5.

just one?
any? the one I
like?